# Biopharmaceutical Analysis and Cell Culture

## Science

May, 2026



Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

# Biopharmaceutical Analysis and Cell Culture (A00841)

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| **Short Title:** | Biopharma. & Cell Culture | | |
| **Faculty** | Science and Computing | | |
| **Department** | Science | | |
| **Cluster** | Biology | | |
| **Author** | KGRENNAN | | |
| **Credits** | 10 | **Level:** | Postgraduate |

## Description of Module / Aims

This module will provide students with an in-depth knowledge and skills in biopharmaceutical production, analysis and cell culture-related topics. Students will be exposed to a set of technologies that provide state-of--the-art tools for genomics, proteomics, metabolomics, drug discovery, screening and the analysis of biomolecules, such as DNA, proteins and small metabolites (biopharmaceuticals). In addition, the module will also introduce different chromatographic and spectroscopic techniques to analyse biopharmaceuticals, as well as other methods ranging from traditional immunoassays to state-of-the-art developments in bioarrays and nano-technology. The practical aspects of the module will cover biopharmaceutical analysis techniques and essential cell culture skills. The course will also train the student to critically evaluate approaches for method development in biopharmaceutical analysis and cell culture, and to critically appraise scientific literature and identity future trends in the area of biopharmaceutical analysis. In summary, the module will furnish the students with a competency required to undertake mammalian cell culture projects, and the ability to select the best techniques needed to analyse different biopharmaceuticals.

## Programmes

| | | |
|---|---|---|
| PHAR-0014 | Certificate in Biopharmaceutical Analysis & Cell Culture (WD_SBIOP_MA) | 5 / 0 / M |
| PHAR-0014 | MSc in Analytical Science with Quality Management (WD_SANSC_R) | 1 / 0 / E |
| PHAR-0014 | PG Dip in Analytical Science with Quality Management (WD_SANSC_G) | 1 / 0 / E |

## Indicative Content

- Biopharmaceutical products: biologics; immunotherapies; vaccines; stem-cell therapeutics; gene therapies; biosimilars; biosuperiors
- Core concepts in molecular biology and recombinant protein production: gene requirements; transfection and cloning; prokaryotic and eukaryotic cell culture systems
- Optimisation of biopharmaceutical product and bioprocessing; upstream and downstream bioprocessing; screening and selection of cells; cell line engineering; metabolic engineering; post-translational modifications
- Case studies of recent biopharmaceutical products & emerging technologies/topics in biopharmaceutical production (transgenics, cell line engineering, designer peptides etc.)
- Ethics and public concern over biopharmaceutical production from genetically modified organisms
- Introduction to mammalian cell culture; history, perspectives and biology of cultured cells
- Sources of mammalian cells & types available; selection of appropriate media, supplements and growth conditions for different cell lines; bioreactors
- Importance of sterility, good culture practice (aseptic techniques), clean room and environmental control; sources of biocontamination and detection and elimination; disposal of cells and health & safety in cell culture lab
- Generation and maintenance of cell banks; maintenance and sub-culturing of cell lines; cell counting and viability checks

- Proteomics in biopharmaceutical analysis and quality control: protein/peptide and post-translational modification analysis using spectroscopic, chromatographic, electrophoretic and other analytical techniques e.g. capillary electrophoresis, SDS-PAGE, Western blot, advanced mass spectroscopy, ion chromatography and nuclear magnetic resonance spectroscopy
- Bioassays and immunoassays in biopharmaceutical analysis and quality control: cell-based assays, immunoassays, flow cytometry, cytometric bead arrays, binding assays and surface plasmon resonance
- Genomics in biopharmaceutical analysis and quality control: DNA sequencing, micro-array analysis and real-time PCR

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Evaluate the principles of biopharmaceutical analysis and cell culture.
2. Evaluate the different techniques (traditional and modern) used in biopharmaceutical analysis and cell culture.
3. Design suitable practical techniques for biopharmaceutical analysis from troubleshooting to successful experimental outcomes.
4. Determine the appropriate medium and growing conditions for different mammalian cells and manage routine maintenance and troubleshooting of the cells.
5. Appraise approaches for method developments and future trends in the area of biopharmaceutical analysis and cell culture through scientific literature.
6. Comment on current and emerging biopharmaceutical products, trials and topics such as ethics and public concern.

## Learning and Teaching Methods

- Lectures.
- Laboratory practical.
- Independent learning.
- This module will use a blended learning approach combining lectures with activities, forums, webinars and videos delivered through the Moodle virtual e-learning platform.

## Assessment Methods

| | Weighing | Outcomes Assessed |
|---|---|---|
| Continuous Assessment | 100% | |
| *In-Class Assessment* | 25% | 1,2, |
| *Assignment* | 30% | 6 |
| *Lab Report* | 40% | 3,4 |
| *Participation* | 5% | 5 |

## Assessment Criteria

**<40%:** Little or no demonstration of an understanding of the fundamentals of biopharmaceutical analysis and cell culture.

**40%-59%:** Ability to show limited understanding of the fundamentals of biopharmaceutical analysis and cell culture. May have some difficulty with applying the theory. Will display some ability with associated demonstrations as well as basic critical thinking and problem-solving skills.

**60%-69%:** As well as the above, ability to show reasonable understanding of the complexities of biopharmaceutical analysis and cell culture, and predict some trends in analytical results. Will display reasonable competence with associated laboratory demonstrations as well as good critical thinking and problem-solving skills.

**70%-100%:** All the above to an excellent level. In addition, will be able to demonstrate comprehensive knowledge and understanding of biopharmaceutical analysis and cell culture-related issues. Should also have the ability to critically evaluate and relate topics in cell culture with biopharmaceutical production and analysis.

## Learning Modes

| Learning Mode | F/T Hours | P/T Hours |
|---|---|---|
| Lecture | | 30 |
| Practical | | 18 |
| Independent Learning | | 222 |

## Essential Material

- Ceccherini-Nelli , L. and B. Matteoli. _Biomedical tissue culture_. Rijeka, Croatia: InTech, 2012.
- Ganapathy, S. and Ganapathy. _Biopharmaceutical Production Technology_. Germany: Wiley-VCH, 2012.
- Tyagi, R. _Cell Biotechnology_. New Delhi, India: Discovery Publishing Pvt.Ltd, 2005.
- Walsh, G. _Biopharmaceuticals: Biochemistry and Biotechnology_. 2nd. Chichester, UK: John Wiley & Sons, 2003.

## Supplementary Material

- "American Type Culture Collection." www.atcc.org
- "BioProcess International." http://www.bioprocessintl.com/
- "Bioanalysis Zone." http://www.bioanalysis-zone.com/
- "Gibco Cell Culture Basics." https://www.thermofisher.com/ie/en/home/references/gibco-cell-culture-basics.html
- "Science Direct." http://www.sciencedirect.com/
- "Web of Science." http://thomsonreuters.com/thomson-reuters-web-of-science/
- Crommelin, D.J.A., R.D. Sindelar and B. Meibohm. _Pharmaceutical Biotechnology: Fundamentals and Applications_. 4th ed. New York, USA: Springer, 2013.
- Helgason, C.D. and C. Miller. _Basic cell culture protocols_. 3rd. Jersey, USA: Humana Press, 2012.
- Miller, J., J.M. Miller and J.B. Crowther. _Analytical chemistry in a GMP environment: a practical guide_. New York, USA: John Wiley & Sons, 2000.
- Skoog, D.A. _Fundamentals of Analytical Chemistry_. 9th ed. Belmont, CA, USA: Thomson-Brooks/Cole, 2013.

## Requested Resources

- Room Type: Biology Lab